

CLEVERCREST AI

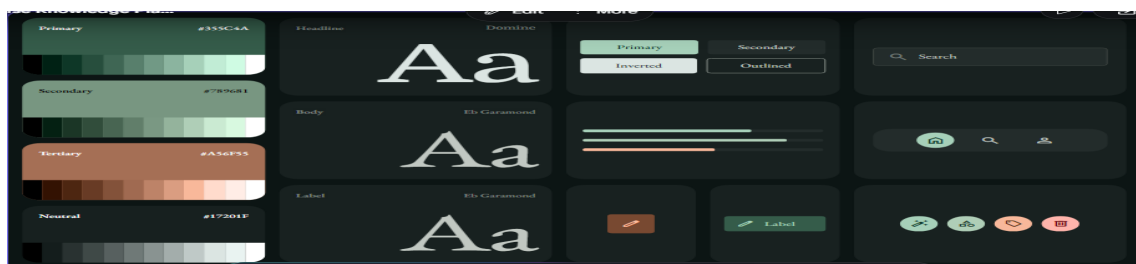
Version 1.0 MVP

Complete Module Report

Multi-Tenant Enterprise Knowledge Intelligence Platform

Foundation • RBAC • Knowledge • Governance • RAG • AI • Utilities

Project Documentation / Semester MVP



Project Type: Multi-Tenant Enterprise Knowledge Intelligence Platform

Primary Development Stack: React + TypeScript, FastAPI + Python, PostgreSQL, ChromaDB, OpenRouter

Development Approach: MVP-first, modular, secure, extension-ready

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The report documents the complete Version 1.0 MVP module scope, user flows, architecture, security model, current implementation foundation, UI/UX direction and future boundaries.

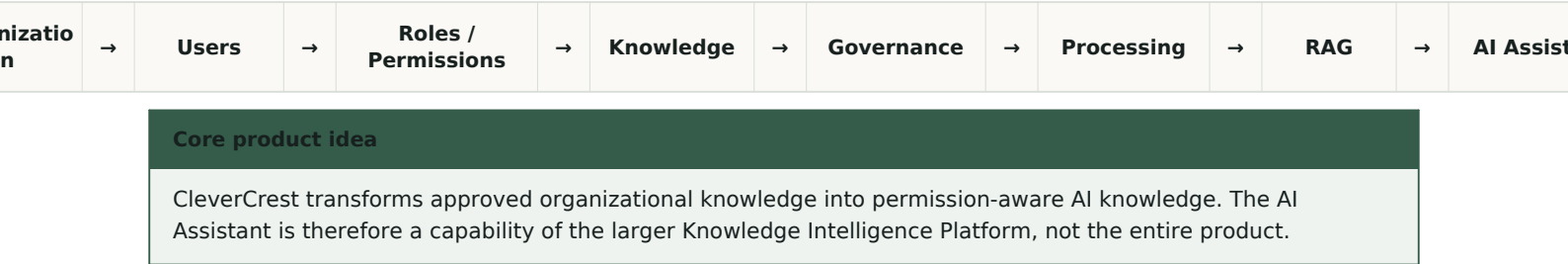
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1. Project Overview

CleverCrest AI is a multi-tenant organizational knowledge intelligence platform. Its purpose is to allow an organization to manage users, roles, knowledge and documents, govern what becomes trusted AI knowledge, and provide authorized users with grounded answers through an AI Assistant.

The product is intentionally broader than a chatbot. The core architecture is:



2. Version 1.0 MVP Objective

Version 1.0 is the semester MVP. The objective is to demonstrate a complete working organizational knowledge lifecycle with appropriate security boundaries.

The V1 scope consists of:

- Authentication
- Multi-tenancy
- Dynamic RBAC
- Knowledge management
- Document governance
- Document processing
- Vector retrieval
- AI assistance
- Access requests
- Notifications
- Audit logs
- Dashboard
- Search

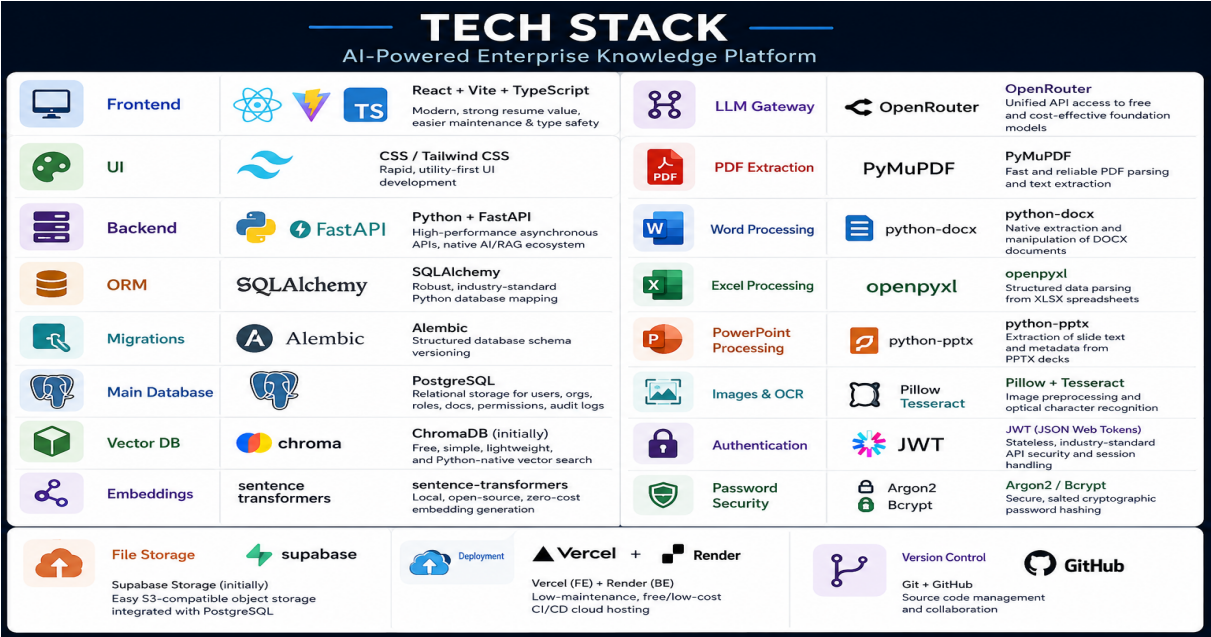
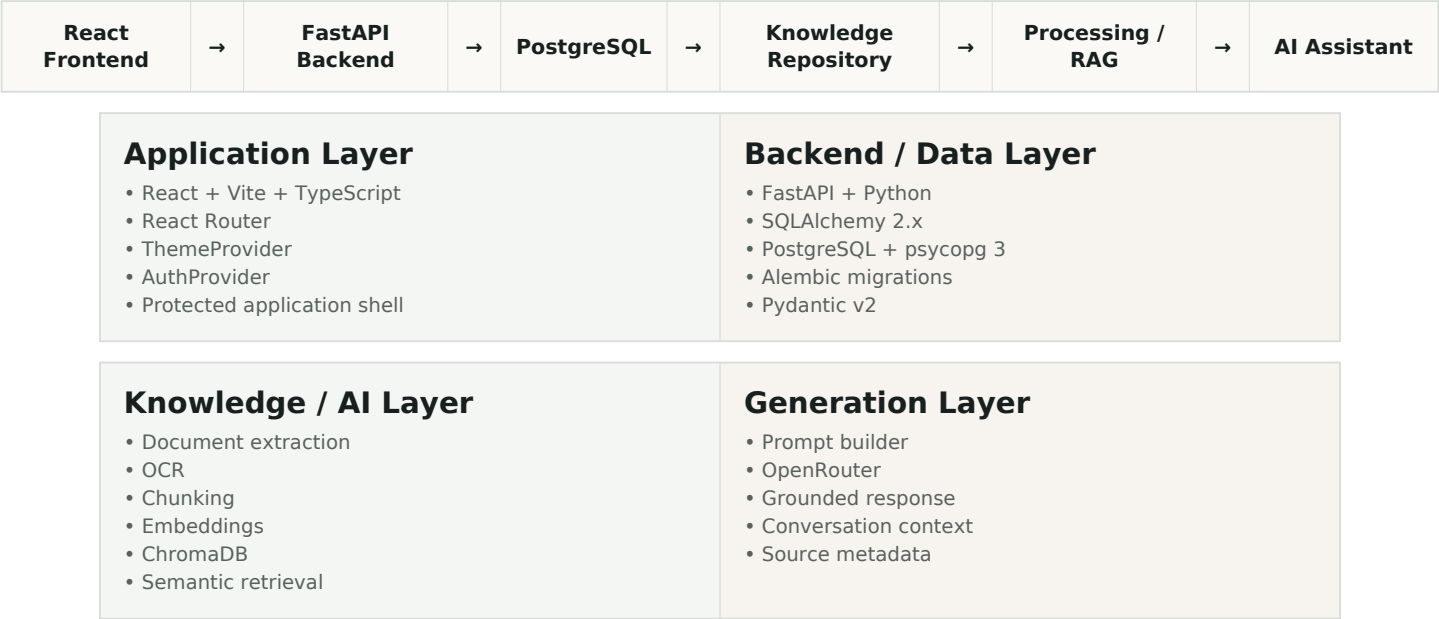
Important scope principle

V1 is not intended to contain every possible AI feature. Advanced features such as multi-agent AI, voice AI, mobile applications, advanced analytics, SaaS billing and large-scale integrations are intentionally deferred.

The implementation strategy is therefore MVP-first: establish identity, tenant isolation, authorization and controlled knowledge first, then build the RAG and AI layers on top of those contracts.

3. Overall V1 Architecture

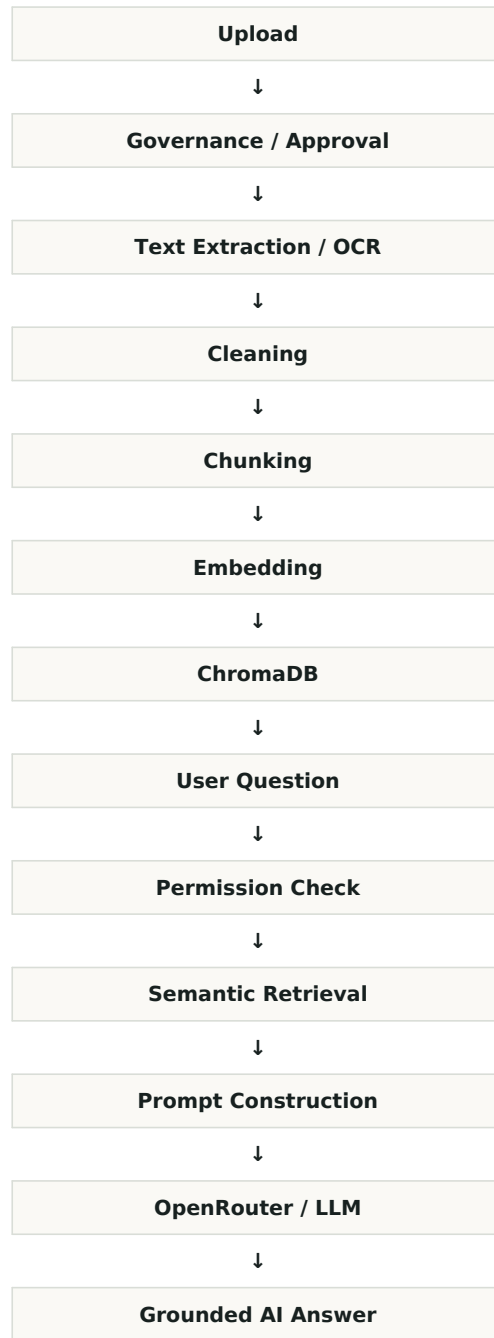
The architecture separates the user interface, backend services, relational data, knowledge processing and AI retrieval/generation layers.



Technology-stack reference supplied for the project.

4. Complete Knowledge-to-AI Lifecycle

This is the most important workflow in CleverCrest. Uploaded knowledge does not immediately become AI knowledge. It must first pass the configured governance workflow.



Core lifecycle rule

Only approved, processed and authorized knowledge should be supplied to the AI model. Retrieval is permission-aware before prompt construction.

5. Current Development Status

The current project already contains a substantial authentication and security foundation. The remaining modules are part of the complete Version 1.0 MVP scope and are not all implemented yet.

Backend	FastAPI, Python, SQLAlchemy 2.x, PostgreSQL, psycpg 3, Alembic, Pydantic v2, PyJWT, Argon2/password hashing, HTTPX.
Frontend	React, Vite, TypeScript, React Router, CSS/design tokens, light/dark theme architecture.
Authentication	Registration, login, email verification, forgot password, reset password, /auth/me, logout, protected route, AuthContext/AuthProvider, HttpOnly JWT cookie.
Database foundation	Organization, User, OrganizationMembership, EmailVerificationToken, PasswordResetToken and OrganizationInvitation.
Email delivery	Gmail SMTP development delivery.
Testing	Latest recorded backend suite: 57 tests passing.
Later V1 modules	RBAC, knowledge repository, governance, processing, vector/RAG, AI Assistant and utilities remain the broader V1 build scope.

Reporting principle

This report distinguishes implemented foundations from V1 modules that are still planned or under development. A module being documented here does not mean its implementation is complete.

6. Module 1 — Authentication

1 Authentication

Implemented

Purpose: Provide secure identity and session management.

V1 scope / responsibilities:

- Registration
- Login
- JWT authentication
- Password hashing
- Logout
- Forgot/reset password
- Email verification
- Basic session handling

User / system flow:

Create Organization	→	Create User	→	Owner Membership	→	Verify Email	→	Login	→	HttpOnly Cookie
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Implementation approach: FastAPI authentication routes connect to PostgreSQL-backed user data.

Passwords use Argon2. JWT is stored in an HttpOnly cookie and the frontend maintains authentication state through AuthContext/AuthProvider.

Security / control point: The JWT contains only user subject and timing claims. One-time email/password-reset tokens are hashed before storage.

2 Organization / Multi-Tenancy

Foundation implemented

Purpose: Establish the tenant boundary for each organization.

V1 scope / responsibilities:

- Organization creation
- Organization identity
- OrganizationMembership
- Tenant-aware backend context
- Support for users belonging to multiple organizations

User / system flow:

Register	→	Create Organization	→	Create User	→	Create Membership	→	Resolve Tenant Context	→	Operate
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Implementation approach: OrganizationMembership is modeled separately from User. The current architecture deliberately keeps organization_id out of the JWT and resolves organization context from membership data.

Security / control point: Tenant isolation must be enforced server-side.

3 Department Management

V1 planned

Purpose: Organize users and knowledge by functional area.

V1 scope / responsibilities:

- Department CRUD
- Description
- Department head/manager
- Department membership

User / system flow:

Admin	→	Create Department	→	Assign Manager	→	Assign Members	→	Use in Knowledge Management
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Implementation approach: Departments are planned as organization-scoped entities connected to users and knowledge metadata.

Security / control point: Department ownership and membership must be checked by backend authorization.

4

User Management

Foundation implemented

Purpose: Allow authorized administrators to control organization membership.

V1 scope / responsibilities:

- Invite user
- Activate/deactivate
- Suspend
- Delete
- Profile management
- Search/filter

User / system flow:

n	→	Users	→	Invite / Select	→	Membership	→	Role	→	Activate	→	User
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Implementation approach: The User and OrganizationMembership foundation is already present. The complete administrative lifecycle remains part of V1.

Security / control point: Only authorized organization administrators should modify membership or status.

5

Role Management

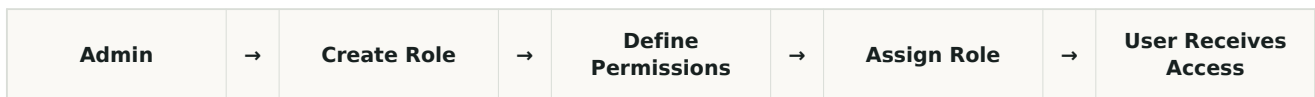
V1 planned

Purpose: Allow organizations to define roles that match their own structure.

V1 scope / responsibilities:

- Organization-defined/custom roles
- Role name and description
- Permission assignment
- User role assignment

User / system flow:



Implementation approach: V1 uses dynamic roles rather than a fixed list of business roles.

Security / control point: Role assignment is an authorization operation and must be enforced by backend APIs.

11. Module 6 — Permission Engine

6

Permission Engine

V1 planned

Purpose: Determine whether a user is allowed to perform a protected action.

V1 scope / responsibilities:

- Upload documents
- Approve/reject
- Delete documents
- Manage collections
- Manage users/departments
- View audit logs
- Manage notification/AI settings

User / system flow:

Request	→	Authenticate	→	Resolve Organization	→	Check Permission	→	Allow / Reject
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Implementation approach: The permission layer sits between authenticated identity and protected resources. The UI may hide unavailable actions, but the backend remains the source of truth.

Security / control point: UI restrictions are not security; protected APIs must enforce authorization.

7

Access Requests

V1 planned

Purpose: Allow users to request access when required knowledge is outside their current permissions.

V1 scope / responsibilities:

- Create request
- Identify resource/collection
- Reviewer decision
- Store decision
- Re-evaluate permissions

User / system flow:

Question	→	Access Missing	→	Submit Request	→	Reviewer	→	Approve / Reject	→	Re-evaluate
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Implementation approach: Access requests connect RBAC, governance and AI retrieval. A decision should affect subsequent permission evaluation.

Security / control point: An access request must never bypass tenant boundaries.

8

Collections

V1 planned

Purpose: Organize knowledge into meaningful organizational domains.

V1 scope / responsibilities:

- CRUD
- Department association

- Owner
- Description
- Tags
- Status/visibility rules

User / system flow:

Authorized User	→	Create Collection	→	Set Metadata	→	Access Rules	→	Add Documents
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Implementation approach: Collections act as organizational knowledge shelves rather than filesystem folders. A collection does not automatically grant access.

Security / control point: Collection access is checked for both management and AI retrieval.

9 Document Management

V1 planned

Purpose: Manage controlled organizational knowledge assets.

V1 scope / responsibilities:

- PDF
- DOCX
- PPTX
- TXT
- Images
- Excel
- Upload/replace/archive/delete
- Owner/department/collection
- Tags/language/version/expiry/status

User / system flow:

Contributor	→	Upload	→	Select Collection	→	Add Metadata	→	Submit	→	Approval Queue
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Implementation approach: Documents are modeled as controlled knowledge assets with lifecycle metadata. Upload does not automatically make a document AI knowledge.

Security / control point: Confidential files should not be directly exposed unless the user's role permits it.

10 Document Processing Queue

V1 planned

Purpose: Make document processing explicit and observable.

V1 scope / responsibilities:

- Pending
- Processing
- OCR
- Chunking
- Embedding
- Completed
- Failed

- Retry

User / system flow:

Pending	→	Processing	→	OCR / Extract	→	Chunk	→	Embed	→	Completed / Failed
---------	---	------------	---	---------------	---	-------	---	-------	---	--------------------

Implementation approach: Processing is designed as a stateful pipeline. The UI should expose current state and clear failure/retry behavior.

Security / control point: Failed processing must be visible and retryable.

16. Module 11 — Approval Workflow

11

Approval Workflow

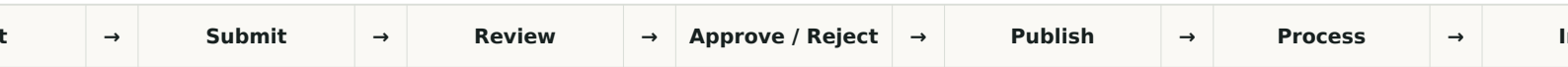
V1 planned

Purpose: Control which information is trusted enough to become AI knowledge.

V1 scope / responsibilities:

- Draft
- Submitted
- Under Review
- Rejected
- Approved
- Published
- Indexed
- Comments
- Revision requests
- Approval timeline

User / system flow:



Implementation approach: Approval is deliberately separated from ingestion. A document can exist in the repository without becoming retrievable AI knowledge.

Security / control point: Only approved and successfully processed knowledge should be available to RAG.

12

Audit Logs

V1 planned

Purpose: Maintain a traceable record of security and knowledge-management activity.

V1 scope / responsibilities:

- Login events
- Upload/delete
- Approve/reject
- Role changes
- Permission changes
- Collection changes
- Important AI configuration changes

User / system flow:



Implementation approach: Important state-changing service operations should create audit records containing actor, organization, action, target and time.

Security / control point: Audit records provide governance traceability.

13 OCR Engine

V1 planned

Purpose: Extract machine-readable text from scanned PDFs and images.

V1 scope / responsibilities:

- Detect OCR requirement
- Extract text
- Preserve basic page/document metadata
- Handle OCR failure

User / system flow:

Document	→	Detect	→	OCR if Needed	→	Extract Text	→	Continue
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Implementation approach: OCR is a processing stage rather than a user-facing AI feature. The planned tooling includes image preprocessing and OCR.

Security / control point: OCR failure moves the job to an explicit failure state.

14 Document Processing & Chunking

V1 planned

Purpose: Normalize extracted content into retrieval-friendly units.

V1 scope / responsibilities:

- Cleaning
- Normalization
- Chunking
- Metadata preservation
- Document-level identifiers

User / system flow:

Extract	→	Clean	→	Normalize	→	Chunk	→	Attach Metadata
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Implementation approach: Raw documents are transformed into normalized chunks while retaining document, collection and organization identifiers.

Security / control point: Chunk metadata must preserve tenant and document identity for secure retrieval.

15 Embedding Engine

V1 planned

Purpose: Convert document chunks and user queries into numerical vectors.

V1 scope / responsibilities:

- Open-source embedding model
- Generate embeddings
- Attach metadata
- Update when content changes
- Delete when knowledge is removed

User / system flow:

Text Chunk	→	Embedding Model	→	Vector + Metadata	→	Store / Update / Delete
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Implementation approach: The planned approach uses a local/open-source sentence-transformers embedding model.

Security / control point: Embeddings must be updated or removed when their source knowledge changes.

21. Module 16 — Vector Database

16

Vector Database

V1 planned

Purpose: Store embeddings and enable similarity search.

V1 scope / responsibilities:

- ChromaDB collections/indexes
- Vector insert/update/delete
- Top-K similarity search
- Metadata filtering

User / system flow:

Embedding	→	ChromaDB	→	Top-K Search	→	Metadata Filters	→	Relevant Chunks
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Implementation approach: ChromaDB is the planned lightweight vector store for the V1 MVP. Metadata is stored with vectors so retrieval can enforce organization and permission constraints.

Security / control point: Similarity alone is never sufficient; metadata filtering is part of the security boundary.

17

Semantic Search

V1 planned

Purpose: Retrieve knowledge by meaning rather than exact keyword matching.

V1 scope / responsibilities:

- Query embedding
- Top-K retrieval
- Collection filtering
- Tenant filtering
- Permission filtering
- Basic ranking

User / system flow:

Question	→	Query Embedding	→	Tenant Filter	→	Permission Filter	→	Collection Filter	→	Similarity Search	→	Top-K Results
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Implementation approach: The retrieval service applies organization and permission constraints before constructing context for the LLM.

Security / control point: Permission filtering occurs before context construction.

18

Prompt Builder

V1 planned

Purpose: Construct a grounded LLM request from authorized context.

V1 scope / responsibilities:

- User question
- Conversation context
- Retrieved chunks
- Organization/system instructions

- Response constraints

User / system flow:

Question	→	Authorized Context	→	Conversation	→	Instructions	→	Prompt
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Implementation approach: The prompt builder combines the current question, bounded recent conversation context, organization instructions and retrieved chunks.

Security / control point: It receives already-filtered context and must not independently discover unauthorized content.

19

AI Response Engine

V1 planned

Purpose: Generate grounded responses using retrieved organizational knowledge.

V1 scope / responsibilities:

- OpenRouter integration
- Low-cost/free model support
- Response generation
- Streaming where practical
- Source metadata
- Basic confidence/quality indicator

User / system flow:

Authorized Prompt	→	OpenRouter	→	LLM	→	Response	→	Metadata
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Implementation approach: OpenRouter is the planned provider gateway so the model/provider can be changed without redesigning the rest of the knowledge pipeline.

Security / control point: Generated answers should be grounded in approved authorized knowledge.

20

AI Assistant / Chat

V1 planned

Purpose: Provide the end-user interaction layer over the knowledge engine.

V1 scope / responsibilities:

- New conversation
- Questions and answers
- Follow-up questions
- Rename/delete chat
- Chat history

User / system flow:

Assistant	→	New Conversation	→	Ask	→	Retrieve	→	Answer	→	Follow-up	→	H
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Implementation approach: The Assistant is designed as an organizational knowledge interface rather than a generic chatbot clone.

Security / control point: Chat requests inherit authenticated user and organization context.

26. Module 21 — Conversation Memory

21

Conversation Memory

V1 planned

Purpose: Maintain enough recent context for natural follow-up questions.

V1 scope / responsibilities:

- Session context
- Previous turns
- Conversation identifiers
- Controlled context window

User / system flow:

Question 1	→	Answer	→	Question 2	→	Reuse Recent Context	→	Answer
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Implementation approach: Conversation turns are associated with a conversation identifier and bounded context is passed to the prompt builder.

Security / control point: Conversation history must remain tenant/user scoped.

22

Permission-Aware AI

V1 planned

Purpose: Prevent unauthorized knowledge retrieval before information reaches the LLM.

V1 scope / responsibilities:

- Tenant check
- Permission check
- Collection/document filtering
- Access request when needed
- Decision recording

User / system flow:

Question	→	User + Tenant	→	Permissions	→	Allowed Knowledge	→	Retrieve	→	Answer / Request
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Implementation approach: This is the central security behavior of the AI layer: tenant and permission filters are applied before context reaches the model.

Security / control point: Unauthorized source content must never be sent to the LLM.

23

AI Response Metadata

V1 planned

Purpose: Provide limited transparency without exposing confidential source contents.

V1 scope / responsibilities:

- Approved organizational knowledge basis
- Non-sensitive source title/version metadata
- Basic confidence indicator

User / system flow:



Implementation approach: The interface can expose document title/version metadata while avoiding confidential document previews.

Security / control point: Metadata itself must be filtered for sensitivity.

24 Notifications

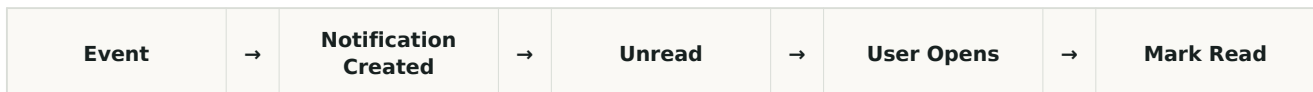
V1 planned

Purpose: Inform users about important workflow events.

V1 scope / responsibilities:

- Approval/rejection
- Access granted/rejected
- Processing completed
- System notifications
- Unread state

User / system flow:



Implementation approach: Notifications are intended to be driven by important backend events. Gmail SMTP can support selected email notifications.

Security / control point: Notification visibility must respect user and organization ownership.

25 Dashboard

V1 planned

Purpose: Provide an operational overview for management users.

V1 scope / responsibilities:

- Document count
- Collection count
- User count
- Pending approvals
- Processing status
- Basic AI usage indicator

User / system flow:



Implementation approach: The dashboard is designed as an editorial operational overview rather than a wall of colorful KPI cards.

Security / control point: Dashboard queries must aggregate only current-organization data.

26 Management Search

V1 planned

Purpose: Help management users locate platform objects directly.

V1 scope / responsibilities:

- Document search

- Collection search
- User search
- Role search
- Filtering

User / system flow:

Search	→	Object Type	→	Filter	→	Results	→	Open Authorized Object
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Implementation approach: Management-side search supports direct discovery of platform objects while ordinary end users primarily use the AI Assistant for knowledge discovery.

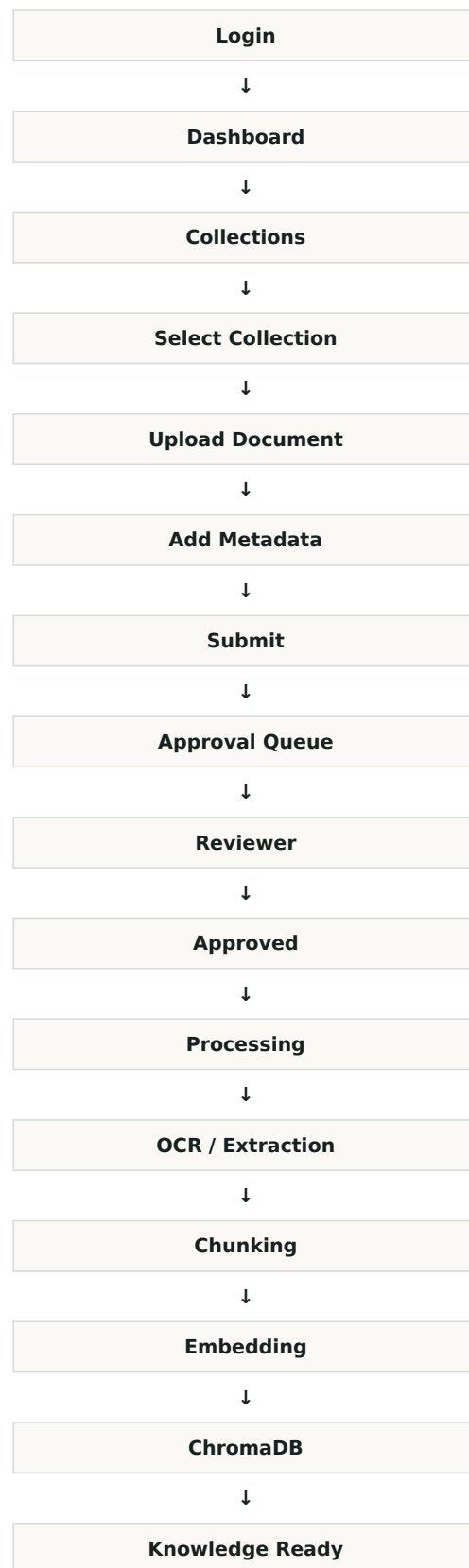
Security / control point: Search results use the same tenant and permission checks as direct object access.

32. Complete Admin User Flow

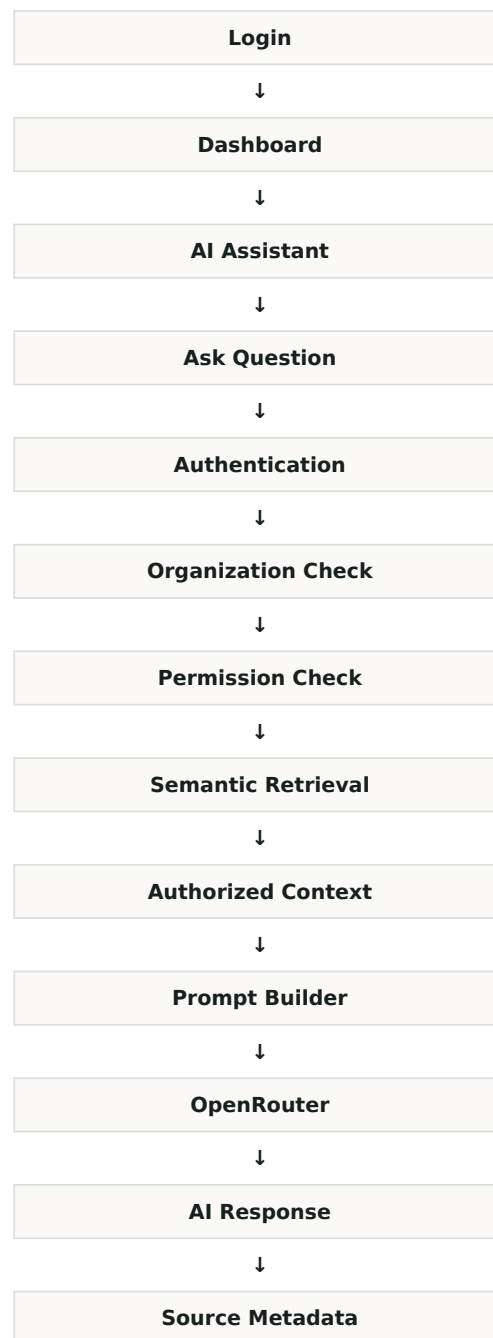


The administrator is the primary management actor. The flow begins with authenticated access and continues through organization structure, users, authorization, knowledge management, governance and operational monitoring.

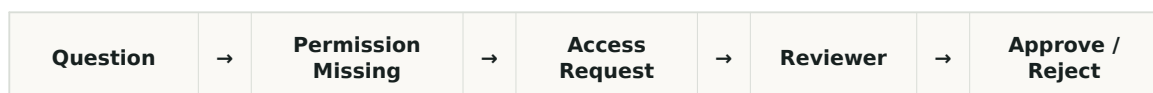
33. Complete Knowledge Contributor Flow



34. Complete Normal User Flow



If access is unavailable, the flow branches to an access request rather than bypassing the authorization boundary.



35. Complete Document Lifecycle



Lifecycle distinction

Uploaded ≠ Approved ≠ Indexed ≠ Available to every user. Each state has a separate purpose and security implication.

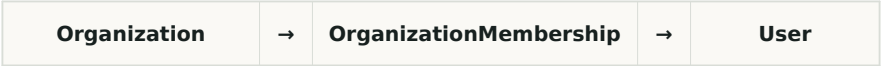
36. Security Architecture

CleverCrest's security model is layered. Each layer addresses a different boundary: identity, tenant context, authorization, knowledge governance, retrieval and auditability.

Authentication	→	Organization	→	Authorization	→	Governance	→	RAG Security	→	Audit
Authentication		JWT authentication in an HttpOnly cookie; Argon2 password hashing.								
Organization		OrganizationMembership establishes the tenant relationship.								
Authorization		Roles and permissions determine protected actions.								
Governance		Documents pass through approval before becoming AI knowledge.								
RAG Security		Tenant, permission and collection filters are applied before context reaches the LLM.								
Audit		Important actions are intended to create traceable audit events.								
Security principle										
The security boundary is not the chat UI. It is the backend: identity → tenant context → permission checks → approved knowledge → filtered retrieval → grounded generation.										

37. Database Foundation

The current backend database architecture uses PostgreSQL. The model foundation deliberately separates user identity from organization membership so that a user can belong to multiple organizations.



Organization	Tenant boundary.
User	Identity and account.
OrganizationMembership	Relationship between user and organization.
EmailVerificationToken	One-time email verification.
PasswordResetToken	One-time password recovery.
OrganizationInvitation	Organization membership invitation foundation.

V1 future domain entities: Departments, Roles, Permissions, Collections, Documents, Approvals, Audit Logs, Notifications and Conversations.

38. Frontend Architecture



Public routes	Login, Register, Verify Email, Forgot Password, Reset Password.
ThemeProvider	Light/dark theme and design token architecture.
AuthProvider	Central authentication/session state.
Protected App	Application shell available after authentication.
Credentialed requests	Frontend sends credentials so the HttpOnly session cookie is used by the browser.

39. Current Authentication Routes

POST	/auth/register	Create organization/user foundation and verification flow.
POST	/auth/login	Authenticate user and create session cookie.
GET	/auth/verify-email	Verify email using one-time token.
POST	/auth/forgot-password	Start password recovery with generic response.
POST	/auth/reset-password	Set new password using reset token.
GET	/auth/me	Return current authenticated user/session information.
POST	/auth/logout	End the current session.

40. Current Testing Status

Latest recorded result

57 backend tests passed.

The current backend test foundation covers:

- Registration
- Login
- Email verification
- Password reset
- /auth/me
- Logout
- Security utilities
- Token functionality
- Email service

There have also been non-failing warnings around HTTP client/dependency deprecations and database connection cleanup. These are warnings rather than failed tests.

41. UI/UX Design System

CleverCrest uses the Crested Intelligence visual concept. The design goal is: quiet confidence + intelligent structure + controlled knowledge.

Personality	Intelligent but not futuristic.
Positioning	Premium but not luxurious.
Technical tone	Technical but approachable.
Security tone	Secure but not intimidating.
Visual rhythm	Calm but not boring.
Structure	Structured but not rigid.

Main visual language:

Ink	#17201F
Deep Moss	#355C4A
Sage	#789681
Mineral	#F3F1EA
Bone	#FAF9F5
Graphite	#59615E
Clay	#A56F55
Mist	#D9DDD7
Critical	#B9534B
Success	#4F765E

42. Typography

Primary font: Plus Jakarta Sans

Page Title	28-34 px
Section Title	20-24 px
Card Title	15-18 px
Body	14-15 px
Metadata	12-13 px

The design system also specifies generous line height for AI responses so that long-form knowledge answers remain comfortable to read.

43. AI Assistant UI Philosophy

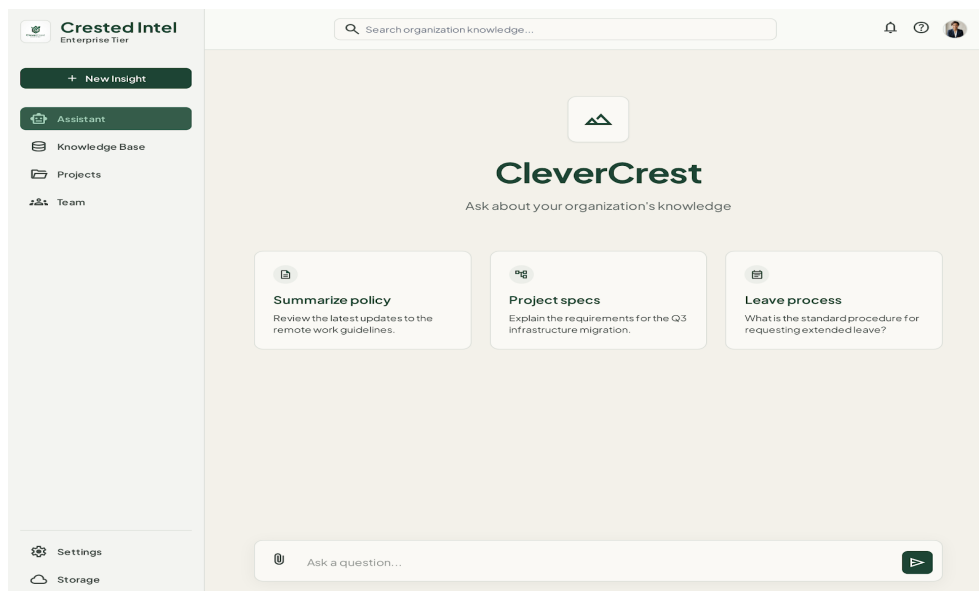
The AI Assistant should not look like a normal ChatGPT clone. The intended direction is an intelligent organizational knowledge system.

Empty-state concept

CleverCrest Ask about your organization's knowledge. [Ask CleverCrest anything...]

Suggested questions can include:

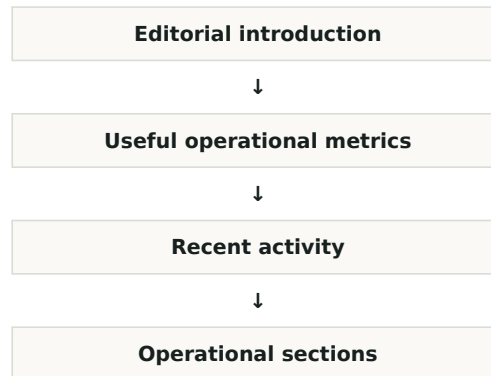
- Summarize the latest policy
- Explain the project requirements
- What is the process for requesting leave?
- Generate questions from available knowledge
- Explain this topic in simple terms



Supplied light-theme AI Assistant design reference.

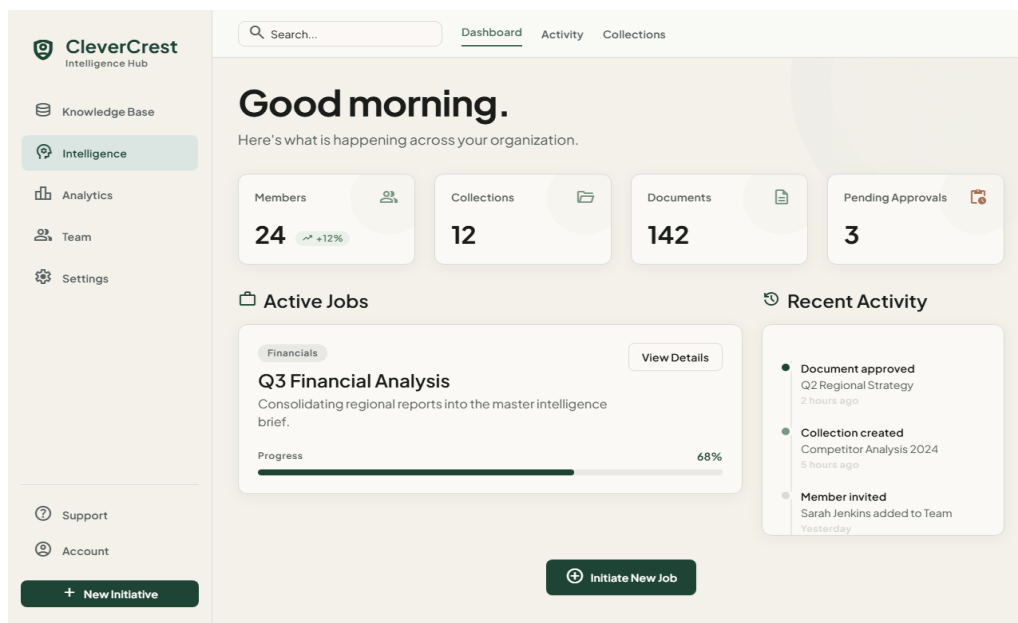
44. Dashboard UI Philosophy

The dashboard should provide an operational overview rather than becoming a wall of colorful KPI cards.



Recommended operational metrics:

- Members
- Collections
- Documents
- Pending Approvals
- Processing Jobs
- AI Conversations



Supplied light-theme dashboard design reference.

45. Document UI Philosophy

Documents should appear as controlled knowledge assets rather than a generic file-drive interface.

File type	Document format.
Document name	Knowledge asset name.
Collection	Organizational knowledge shelf.
Uploader	Contributor identity.
Status	Lifecycle state.
Version	Knowledge version.
Updated time	Latest modification.
Actions	Actions available to the user's role.

46. Upload UI

Controlled knowledge intake

Add knowledge Drop files here or browse PDF · DOCX · PPTX · XLSX · Images Collection Description [Submit for approval]

Technical concepts such as embeddings, vector databases, RAG and chunking are intentionally hidden from normal users. The interface communicates the business action: adding controlled knowledge.

47. Approval Queue UI

The approval interface is a governance console.

Queue count	Number of pending items.
Document metadata	Name, collection, uploader and version.
Submission time	When the item entered review.
Review history	Previous review events and reviewer.
Decision	Approve or reject.

48. Processing UI

Processing should visually represent the knowledge pipeline instead of showing only a generic spinner.



Semantic states such as Processing, Failed, Retry and Completed make background processing understandable to users.

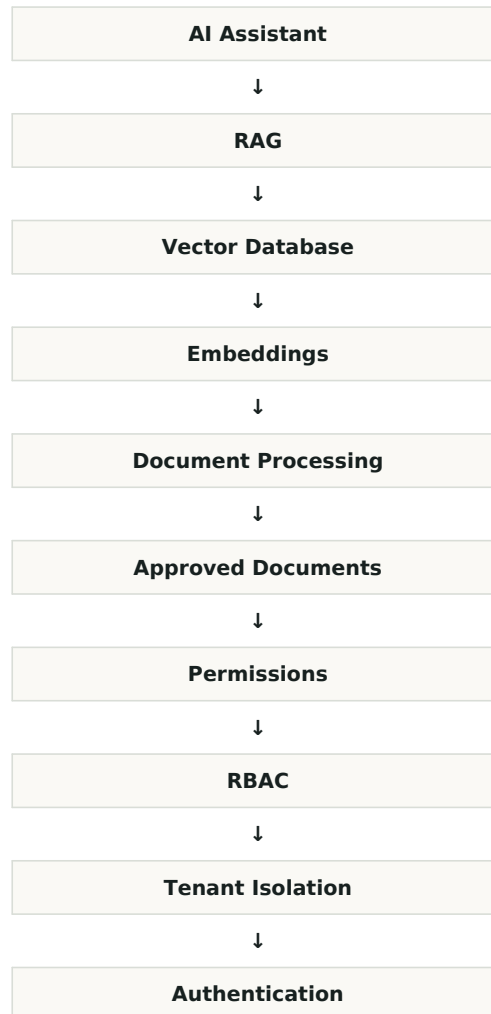
49. V1 Module Dependency Order

1	Project Foundation
2	Authentication
3	Multi-tenancy
4	RBAC
5	Collections & Documents
6	Approval Workflow
7	Processing Pipeline
8	Vector Layer
9	RAG Service
10	AI Assistant
11	Utilities
12	Hardening

Tenant	→	RBAC	→	Knowledge	→	Governance	→	Processing	→	Vector	→	RAG	→	AI	→
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50. Why We Should Not Build AI First

The AI Assistant depends on multiple security and knowledge contracts. Building the AI layer first would create a risk of designing retrieval before the authorization and governance boundaries are stable.



Dependency principle

Authentication → tenant isolation → authorization → controlled knowledge → processing → retrieval → AI. The AI layer is the final consumer of these contracts.

51. V1 Completion Criteria

Tenant Isolation	A user can access only the organization to which the account belongs.
RBAC	Unauthorized API actions are rejected.
Knowledge Governance	Unapproved documents cannot be retrieved by AI.
Processing	Documents move through visible processing states and reach indexed or failed status.
RAG	An authorized user receives a grounded response based on relevant approved context.
Access Requests	A user can request access and an authorized reviewer can make the decision.
Auditability	Important actions create audit records.
Usability	Administrators can operate the knowledge workflow without developer intervention.
Deployment	The MVP can run using the selected infrastructure within practical resource limits.

52. Quality Gates

Phase Gate	Core user flow works end-to-end.
Security Gate	Authorization is enforced server-side.
Data Gate	Tenant identifiers and ownership relationships exist where required.
AI Gate	RAG never retrieves knowledge the requesting user is not authorized to use.
Failure Gate	Failed processing is visible and retryable.
Audit Gate	Important state changes are traceable.
Deployment Gate	Environment, secrets, CORS, migrations, storage and model configuration are reproducible.
Documentation Gate	APIs and major architectural decisions are documented.

Definition of done

A module is not considered complete merely because its screen exists. Its core user flow must work end-to-end, protected resources must be server-authorized, and important failure states must be visible.

53. What Is NOT Part of V1

The following capabilities are deliberately deferred from the semester MVP.

Advanced AI

- LangGraph multi-agent system
- HR Agent
- Legal Agent
- Finance Agent
- Training Agent
- Research Agent
- Project Agent

Voice

- Speech input
- Speech-to-text
- Text-to-speech
- Voice assistant

Advanced Intelligence

- Recommendation engine
- Adaptive learning
- Knowledge-gap detection
- Advanced analytics

Advanced Document Features

- Handwriting OCR
- Video ingestion
- Audio ingestion
- Advanced document intelligence

Platform

- Native mobile application
- Real-time collaboration
- SaaS billing
- Large external integrations
- Enterprise SSO/MFA beyond MVP requirements

54. Future Versions

The roadmap intentionally separates the semester MVP from later product expansion.

V2.0 — Enterprise Enhancement	Analytics, knowledge-gap detection, document intelligence, duplicate detection, smart tagging, version comparison, advanced document lifecycle, bookmarks, saved answers, chat export and improved search.
V3.0 — AI Intelligence	Multi-agent system, voice AI, recommendations, adaptive learning, enterprise AI automation and mobile platform.
V4.0 — SaaS Product	Subscription, billing, quotas, payment gateway, platform super admin, tenant management, Microsoft 365, Google Workspace, Slack, Teams, SharePoint, SSO, MFA, encryption/key management and compliance reporting.

Scope boundary

These future capabilities are product directions beyond the semester MVP. They should not be used as completion criteria for Version 1.0.

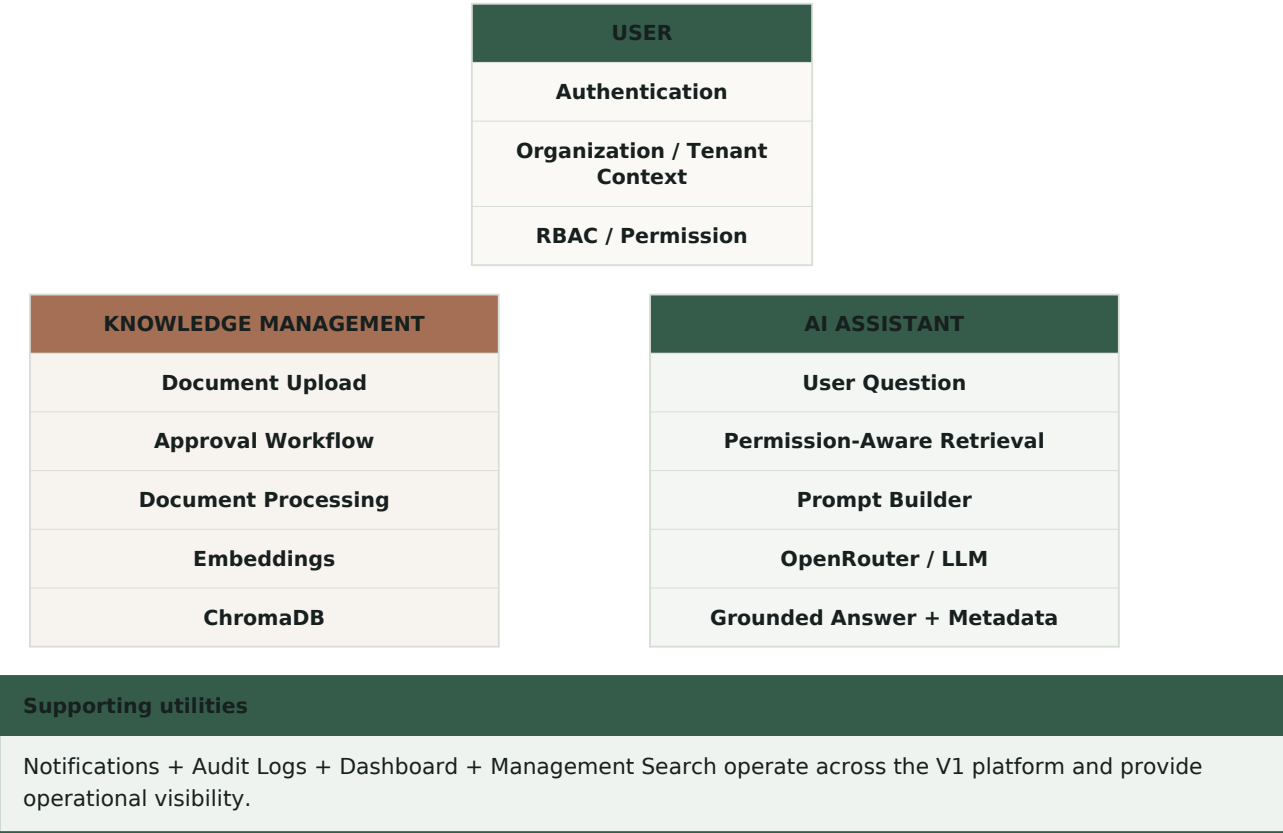
55. Final V1 Module Structure

PLATFORM FOUNDATION	Authentication • Organization • Departments • Users
ENTERPRISE RBAC	Roles • Permissions • Access Requests
KNOWLEDGE REPOSITORY	Collections • Documents • Processing Queue
KNOWLEDGE GOVERNANCE	Approval Workflow • Audit Logs
AI KNOWLEDGE ENGINE	OCR • Processing • Chunking • Embeddings • ChromaDB • Semantic Search • Prompt Builder • AI Response
AI ASSISTANT	Chat • Conversation Memory • Permission-Aware AI • Response Metadata
PLATFORM UTILITIES	Notifications • Dashboard • Management Search

Foundation	→	RBAC	→	Knowledge	→	Governance	→	RAG	→	AI	→	Utilities
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56. Overall V1 Workflow

The following diagram combines the major user, knowledge and AI paths into one overall V1 view.



57. Final Project Position

CleverCrest AI V1 is a secure, multi-tenant organizational knowledge platform in which approved organizational documents are transformed into permission-aware AI knowledge and exposed to authorized users through an intelligent assistant.



Fundamental security rule

No unauthorized organizational knowledge should reach the AI model.

The project is therefore not simply a chatbot application. It is a controlled organizational knowledge lifecycle in which authentication, tenant isolation, authorization, governance and processing establish the conditions under which AI can safely operate.

The Version 1.0 objective is to produce a coherent, secure and accurate MVP demonstrating this real enterprise knowledge lifecycle. Later versions can extend the same architecture with advanced intelligence, automation, integrations, mobile experiences and SaaS capabilities.

End of Version 1.0 MVP Module Report

CleverCrest AI • Semester Project Documentation